

Card 1: Lock-Free B-Link Tree

Sled uses B-Link trees. Node splits introduce right-sibling links, letting readers traverse split boundaries without locks.

Card 5: Concurrency Trade-off

Locked B-Trees block on splits. B-Link uses CAS and sibling links to isolate split conflicts and keep reads lock-free.

Card 9: Page Consolidation

Background threads merge Delta Chains into flat Base Pages, swapping Page Table pointers atomically via CAS.

Card 13: Async Parent Updates

Asynchronously appends Split records to parent nodes using thread pools to avoid blocking active write loops.

Card 2: Page Table CAS EBR

Maps logical Page IDs to Delta Chain pointers. Swapping page pointers atomically via CAS. Managed under EBR.

Card 6: Log-Structured Layout

Appends page writes, updates, and deletes sequentially to active segments. Aligns to SSD write cycles.

Card 10: WAL Group Commits

Buffers active log operations in memory. Group-commits buffered states to disk to minimize seek latency.

Card 14: Split Conflicts Retry

Resolves concurrent splits or writes on same parents via Spin-CAS loops, reloading pointers on failure.

Card 3: Leaf Page Layout

Key-Value slots are stored in a sorted binary-searchable array. Writes append Put deltas onto memory chains.

Card 7: Page Delta Chains

Represents logical page state as a memory chain ending at a flat Base Page. Limits chain lengths to prevent read bloat.

Card 11: Leaf Overflow Split

Splits leaf nodes when data sizes exceed boundaries. Allocates new Page ID and links right-pointers atomically.

Card 15: Crash Recovery Build

Reconstructs Page Tables by scanning log files sequentially from the last checkpoint up to the log end.

Card 4: Inner Node Routing

Stores child Page IDs and boundary keys. Traverses right-links dynamically if child split isn't routed yet.

Card 8: Segment Mapping

Buffer cache reads from Page Table pointers, redirecting to absolute disk LSN sectors if blocks are evicted.

Card 12: Split-Phase Traversal

Allows readers to traverse right links from Leaf A to B lock-free during ongoing split operations.

Card 16: LSN Generation

Tags physical log entries with monotonic 64-bit Log Sequence Numbers to track write history during recovery.

Card 17: Epoch Reclamation EBR

Regulates thread access in memory zones. Defers node deallocations until threads exit associated Epoch contexts.

Card 21: Transaction CAS Loops

Executes ACID transactions optimistically, validating read-write sets and committing via Page Table CAS.

Card 25: Cache Eviction LRU

Manages flat Base Pages in memory, running LRU eviction algorithms when cache memory limits are breached.

Card 18: GC Segment Cleanups

Scans cold segments, copy-evacuates active pages to new active zones, and format empty segments for reuse.

Card 22: Key Range Locks

Enforces range locking during the validation phase to prevent phantom reads during optimistic commits.

Card 26: Failpoint Chaos Test

Embeds Failpoints in page splits and writes, verifying safety during chaos testing.

Card 19: Write Amplification

Separates hot and cold records and coordinates GC activities to reduce drive wear and extend SSD life.

Card 23: Read-Your-Own-Writes

Queries private transaction write buffers before global Page Tables to ensure session consistency.

Card 27: Telemetry Metrics

Exposes metrics on active pages, dirty ratios, EBR queue lengths, and WAL throughput for real-time monitoring.

Card 20: Compaction Thresholds

GC monitors segment stale ratios and active density, triggering compaction when stale ratio exceeds 70%.

Card 24: Concurrent Handles

Implements Send and Sync on DB handles, allowing clones across threads without channel bottlenecks.

Card 28: Performance Tunings

Adjusts segment sizes and compaction thresholds to balance flash wear against database execution speed.